LSP Task1

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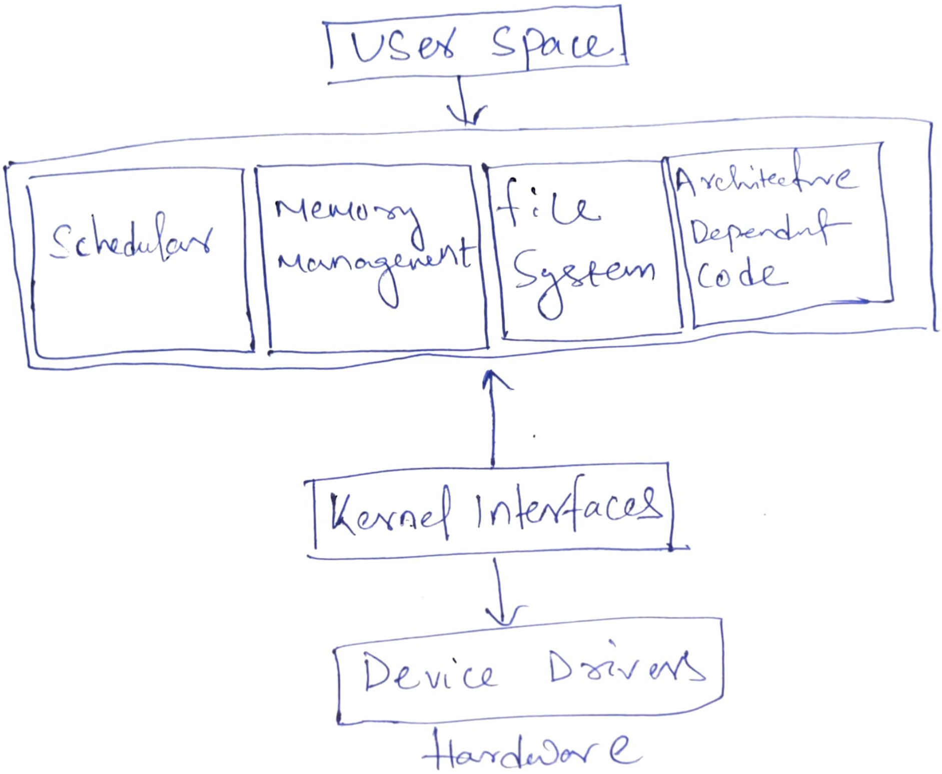
Linux kernel Architecture



i) **Scheduler** – The Schedular is incharge of deviding which ~~powers~~ process runs on the CPU and for how long. It handles process prioritization and convert switching to ensure efficient Multitasking.

ii) **Memory Management** – This component manages how memory is allocated and feed for application and process. It includes mechanisms like virtual memory and ~~Paging~~ Paging to optiming RAM usage.

iii) **file System** The file System organises and controls access to data stored on devices. It handles files reading writing, permission and supports different file format.

iv) Network Stack – This part deals with all networking operations. It enables communication between devices using various protocols like TCP/IP, ensuring data is sent and required properly.

v) Architecture Dependent code :– These are the hardware specific parts of the kernel. They let the same kernel work on different processors and platform by adapting to their unique features.

vi) Kernel Interfaces :– These are the links between user applications and the kernel, including system cells and libraries. They let programs request kernel level.

vii) Device Drivers :– Drivers acts as translator between hardware and the rest of the system. They allows the kernel to communicate with devices like keyboard, drives and network cards.